Gerard Alexander Gibson

Software Developer | Full-Stack, Backend and Mobile Development
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PROFESSIONAL SUMMARY

Final-year Application Development student with practical experience building full-stack web, Android and cross-platform applications using C# and ASP.NET Core, React with Node.js, Kotlin and Flutter. Led a six-member Work-Integrated Learning team and took primary responsibility for the REST API connecting a Flutter mobile app and an ASP.NET Core MVC staff portal. Project work covers secure REST APIs with JWT and role-based access control, SQL Server, MongoDB and Firebase data stores, Microsoft Azure services, automated testing and Git-based collaboration. Seeking a graduate or junior software engineering role.

TECHNICAL SKILLS

Programming Languages: C#, JavaScript, Kotlin, Dart, SQL, Java

Backend Development: ASP.NET Core Web API, ASP.NET Core MVC, Entity Framework Core, Node.js, Express.js, RESTful API design

Frontend Development: React, Vite, HTML, CSS

Mobile Development: Android SDK (Kotlin, XML layouts), Flutter

Databases: Microsoft SQL Server, Azure SQL, MongoDB (Mongoose), Firebase

Cloud and DevOps: Microsoft Azure (Azure SQL, Azure Storage), Docker, Docker Compose, GitHub Actions, Git, GitHub

Testing: xUnit, Jest, Supertest, Vitest, Postman, Newman, Swagger

Security: JWT authentication, bcrypt password hashing, role-based access control, Helmet, rate limiting, input sanitisation

RELEVANT EXPERIENCE

Software Engineering Team Lead (Work-Integrated Learning) February 2026 – Present

Emeris, Nelson Mandela Bay | Client-based academic project, completed as part of the degree

- Led a six-member team developing REGGY, a student registration platform for an institutional client, coordinating responsibilities and milestones across API, website, mobile and database workstreams.
- Took primary responsibility for the shared C# and ASP.NET Core Web API (.NET 10) connecting the Flutter mobile app and the ASP.NET Core MVC staff website.
- Worked on authentication, role-based authorisation, validation, testing and troubleshooting across the integrated system.
- Supported the integration of independently developed web and mobile components using GitHub feature branches and integration testing.
- Gathered client feedback, communicated progress and demonstrated functionality to stakeholders; the client approved the work presented.

SELECTED SOFTWARE ENGINEERING PROJECTS

REGGY — Student Registration Management Platform

Team project | C#, ASP.NET Core Web API, ASP.NET Core MVC, .NET 10, Flutter, Dart, Firebase, Git, GitHub

- Multi-platform system digitising student applications, document submission and staff verification, appointment requests, student-number management, and status notifications.
- Provides staff dashboards, reporting and administrative controls, with centralised validation, identity management and role-based authorisation enforced at API level.

HustleHub+ — Secure Full-Stack Freelancer Marketplace

React 19, Vite, Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, Jest, Supertest, Vitest, Postman, Newman

- Developed a marketplace connecting freelancers and clients, with a React frontend and an Express REST API structured into middleware, controllers and services over MongoDB.
- Implemented JWT authentication, bcrypt password hashing and role-based access for Client, Freelancer and Admin roles, hardened with Helmet headers, rate limiting, input sanitisation and NoSQL-injection protection.
- Built gig management with filtering and pagination, booking workflows, simulated payment records and freelancer income summaries; tested with Jest and Supertest, Vitest, and Postman/Newman.

TechMove — ASP.NET Core Service Management Platform

C#, ASP.NET Core MVC, ASP.NET Core Web API, Entity Framework Core, SQL Server, Docker Compose, xUnit, GitHub Actions, Swagger

- Developed a database-driven application for managing clients, contracts (with PDF uploads) and service requests, using Entity Framework Core migrations on SQL Server.
- Implemented business rules that block service requests against expired contracts, and a separate Web API that converts USD to ZAR using an external exchange-rate API.
- Wrote xUnit tests for business rules, validation and conversion logic; used Docker Compose to run local services and configured GitHub Actions CI.

Impilo — Android Mental Wellness Application

Kotlin, Android SDK, XML, Gradle, GitHub Actions, Android Studio

- Developing a native Android app in Kotlin with XML layouts, providing home, task, goal, profile and AI-chat screens, with UI separated from application logic.
- Built task and goal management; mood check-ins, journaling and breathing exercises are in active development.
- Configured a GitHub Actions workflow that sets up Java 17 and the Gradle wrapper and runs the Android debug build to verify compilation.

Savora — Android Budget Management Application

Kotlin, Android SDK, XML, Firebase, Gradle, Android Studio

- Developed a native Android budgeting app with Firebase user authentication for recording and categorising expenses, creating budgets and tracking savings goals.
- Built dashboards summarising income and expenses, extending features across iterative development cycles.

Claimy — Lecturer Claims Management System

C#, ASP.NET MVC, Entity Framework, Azure SQL, JSON, HTML, CSS

- Developed a claims workflow for lecturers to submit claims with supporting documents, with totals calculated automatically from hours worked and hourly rate.
- Implemented role-based review and approval for Lecturer, Coordinator, Manager and Human Resources users, with claim status tracking.
- Stored account and authentication data in Azure SQL and selected operational records as JSON.

TechNog — Azure-Based E-Commerce Management System

C#, ASP.NET MVC, Azure Storage (Blob, Table, Queue, File), HTML, CSS

- Developed product, customer and order management with administrative dashboards and initial customer-facing pages.
- Built the solution around Azure Storage services in response to a business case for moving away from unreliable on-premises infrastructure.

EDUCATION

IIE Bachelor of Computer and Information Sciences in Application Development

Emeris (formerly IIE Varsity College) | Final year, in progress — February 2024 – November 2026 (Expected)

Higher Certificate in Communication Practices

IIE Varsity College | Completed — February 2023 – November 2023

ADDITIONAL STRENGTHS

Technical leadership • Communication and public speaking • Critical thinking • Collaborative problem-solving